

COGNITIVE ASSESSMENT

Target Age: 14–23 Years

Task 1 — Visual Recall

Skill: Working Memory

Task Description: Display 4 symbols for 5 seconds, then hide them and show 8 symbols. The student selects the 4 symbols they saw previously.

Example: ● ▲ ★ ■

● ◆ ▲ ♥

★ ■ ☀️ 🏠

Student Instruction: “Select the symbols you saw before.”

Interaction: Click / Select

Difficulty: Increase the number of symbols and reduce the display time.

Task 2 — Sequential Memory

Skill: Working Memory

Task Description: Display 4 symbols in a specific order, then hide them and show the same symbols in random order. The student rearranges them into the original sequence.

Example: Original: ★ → ● → ▲ → ■

Random: ■ ▲ ★ ●

Student Instruction: “Put the symbols in the same order you saw before.”

Interaction: Drag&Drop

Difficulty: Increase the number of symbols from 3 to 6.

Task 3 — Remember&Follow

Skill: Working Memory

Task Description: Show 3 simple actions together, then hide the instructions. The student performs the actions in the correct order.

Example: Click the red circle → move it to the box → click the green button.

Student Instruction: “Do what you remember.”

Interaction: Click / Drag&Drop

Difficulty: Increase from 2 to 4 steps.

Task 4 — Pattern Recognition

Skill: Fluid Reasoning

Task Description: Display a visual pattern with one missing element. The student selects the element that completes the pattern.

Example: ● ▲ ● ▲ ● ?

Student Instruction: “Choose what comes next.”

Interaction: Click / Select

Difficulty: Use more complex patterns with 2 or more changing features.

Task 5 — Rule Detection

Skill: Fluid Reasoning

Task Description: Show several examples that follow the same rule. The student identifies what should come next using the rule.

Example: ● → ▲ → ■
▲ → ■ → ★
■ → ★ → ?

Student Instruction: “What comes next?”

Interaction: Click / Select

Difficulty: Use less obvious rules and multiple features.

Task 6 — Problem Solving

Skill: Fluid Reasoning / Problem Solving

Task Description: Present a simple visual problem with several possible actions. The student chooses the action that solves the problem.

Example: A path to the goal is blocked. Three possible routes are shown.

Student Instruction: “Choose the best way to solve the problem.”

Interaction: Click / Select

Difficulty: Increase the number of choices and obstacles.

Task 7 — Rule Switching

Skill: Cognitive Flexibility

Task Description: Ask the student to sort symbols using one rule, then change the rule and ask them to continue using the new rule.

Example: Round 1: Sort by COLOR.
Round 2: Sort by SHAPE.

Student Instruction: “Follow the new rule.”

Interaction: Drag&Drop / Select

Difficulty: Increase the number of items and switch rules more frequently.

Task 8 — Visual Matching

Skill: Visual-Spatial Processing

Task Description: Show a target shape and several similar shapes. The student selects the shape that matches the target.

Example: Target: 

Options:    

Student Instruction: “Find the shape that matches the target.”

Interaction: Click / Select

Difficulty: Use similar shapes and rotations at higher levels.

Task 9 — Mental Rotation

Skill: Visual-Spatial Processing

Task Description: Show a target shape and several rotated shapes. The student selects the option that represents the same shape after rotation.

Example: Target: 

Options:    

Student Instruction: “Choose the shape that is the same as the target.”

Interaction: Click / Select

Difficulty: Increase the complexity and number of rotations.

Task 10 — Spatial Relationships

Skill: Visual-Spatial Processing

Task Description: Show objects in different positions and ask the student to identify their spatial relationship.

Example: ● is above ■.

Where is ■ relative to ●?

Student Instruction: “Choose the correct position.”

Interaction: Click / Select

Difficulty: Start with above/below/left/right, then add diagonal positions.

Task 11 — Spatial Construction

Skill: Visual-Spatial Processing

Task Description: Show an arrangement of shapes, hide it, and ask the student to recreate the same arrangement.

Example: ● ▲

■ ★

Student Instruction: “Recreate the same arrangement.”

Interaction: Drag&Drop

Difficulty: Increase from 2 shapes to 6 shapes.

Task 12 — Visual Search

Skill: Processing Speed / Attention

Task Description: Show a target symbol and a group of symbols. The student finds the target as quickly as possible.

Example: Target: ★

● ▲ ■ ★ ◆ ● ■ ▲ ★

Student Instruction: “Find the target symbol.”

Interaction: Click / Select

Difficulty: Increase the number of symbols and distractors.

Task 13 — Symbol Matching Speed

Skill: Processing Speed

Task Description: Show a target symbol followed by a series of symbols. The student identifies whether the target appears in each set.

Example: Target: ★

Set: ● ▲ ★ ■

Student Instruction: “Does the target appear in the set?”

Interaction: Click: YES / NO

Difficulty: Increase the number of symbols and reduce response time.

Task 14 — Target Detection

Skill: Sustained Attention

Task Description: Display symbols one at a time in sequence. The student clicks only when the target symbol appears.

Example: ● → ▲ → ■ → ★ → ● → ◆ → ★

Student Instruction: “Click when you see ★.”

Interaction: Click

Difficulty: Increase the number of symbols and sequence length.

Task 15 — Visual Attention Over Time

Skill: Sustained Attention

Task Description: Display a continuous stream of symbols. The student monitors the screen and responds whenever a specific target appears.

Example: Show 20–30 symbols one at a time. Target: ★

Student Instruction: “Click every time you see ★.”

Interaction: Click

Difficulty: Increase duration, distractors, and time between targets.

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Task 16 — Selective Attention

Skill: Selective Attention

Task Description: Show symbols with different colors and shapes. The student selects only items that meet two conditions at the same time.

Example: Select only BLUE + LARGE shapes.

Student Instruction: “Select only the shapes that match both rules.”

Interaction: Click / Select

Difficulty: Increase distractors and change the conditions.

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Task 17 — Change the Sorting Rule

Skill: Cognitive Flexibility

Task Description: Ask the student to sort items using one rule, then introduce a new rule and ask them to sort the next items using the new rule.

Example: First rule: COLOR

New rule: SHAPE

Student Instruction: “Sort the items using the new rule.”

Interaction: Drag&Drop

Difficulty: Change rules more often and use less obvious differences.

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Task 18 — Switch Between Rules

Skill: Cognitive Flexibility

Task Description: Display a cue before each item indicating which rule to use. The student switches between rules according to the cue.

Example: COLOR → sort by color

SHAPE → sort by shape

Student Instruction: “Follow the rule shown on the screen.”

Interaction: Click / Drag&Drop

Difficulty: Alternate rules more quickly and increase items.

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Task 19 — Find the Best Route

Skill: Planning&Problem Solving

Task Description: Show a simple map with a starting point, a goal, several routes, and obstacles. The student chooses a successful route.

Example: START → available routes → obstacles → GOAL

Student Instruction: “Find the best route to the goal.”

Interaction: Click / Drag

Difficulty: Increase routes, obstacles, and decision points.

Task 20 — Solve the Problem

Skill: Planning&Problem Solving

Task Description: Present a visual situation where the direct solution does not work. The student selects an action that can solve the problem.

Example: A direct path is blocked and several alternative actions are available.

Student Instruction: “What should you do to solve the problem?”

Interaction: Click / Select

Difficulty: Increase possible actions and steps needed.

Task 21 — Find an Alternative

Skill: Planning / Cognitive Flexibility

Task Description: Give the student a route or solution that becomes unavailable. The student must find another way to reach the goal.

Example: The chosen path becomes blocked. Two alternative paths appear.

Student Instruction: “Find another way to reach the goal.”

Interaction: Click / Drag

Difficulty: Increase obstacles and alternative routes.

Task 22 — Multiple Rules

Skill: Following Rules&Instructions

Task Description: Give the student two rules that must be followed at the same time when selecting items.

Example: Select BLUE shapes, but do NOT select CIRCLES.

Student Instruction: “Follow both rules.”

Interaction: Click / Select

Difficulty: Increase rules and distractors.

Task 23 — Multi-Step Instructions

Skill: Following Rules&Instructions

Task Description: Give the student a sequence of 3–4 actions to complete in the correct order.

Example: Select the blue circle → move it to the box → select the star → click Submit.

Student Instruction: “Complete the steps in the correct order.”

Interaction: Click / Drag&Drop

Difficulty: Increase the number of steps and actions.

Task 24 — Find Information Using Multiple Conditions

Skill: Information Processing

Task Description: Show a simple table containing information such as file name, type, and status. The student finds the item that matches multiple conditions.

Example: File | Type | Status

A | PDF | Complete

B | DOC | Complete

C | PDF | Pending

Find: PDF + Complete

Student Instruction: “Find the item that matches all the conditions.”

Interaction: Click / Select

Difficulty: Increase conditions and information shown.

Task 25 — Choose the Best Option

Skill: Decision Making

Task Description: Present a simple situation with several possible actions. The student selects the option that best solves the situation.

Example: Some information is missing.

Options: Ask for the missing information / Guess / Continue without it.

Student Instruction: “Choose the best option.”

Interaction: Click / Select

Difficulty: Use situations with more choices and relevant factors.